AGA KHAN UNIVERSITY EXAMINATION BOARD

SECONDARY SCHOOL CERTIFICATE

CLASS IX EXAMINATION

MAY 2011

Mathematics Paper II

Time allowed: 2 hours 20 minutes Marks 45

INSTRUCTIONS

Please read the following instructions carefully.

1. Check your name and school information. Sign that it is correct.

I agree that this is my name and school. **Candidate's signature**

2. RUBRIC. There are FOURTEEN questions. Answer ELEVEN questions in all. Choices are specified inside the paper.

3. When answering the questions:

   Read each question carefully.
   Use a black pencil for diagrams. DO NOT use coloured pencils.
   DO NOT use staples, paper clips, glue, correcting fluid or ink erasers.
   Complete your answer in the allocated space only. DO NOT write outside the answer box.

4. The marks for the questions are shown in brackets ( ).

5. You may use a simple calculator if you wish.

(ATTEMPT EITHER PART a OR PART b OF Q.1.)

Q.1. (Total 4 Marks)

a. Prove that $\left[\left(\frac{x^2}{y^2}\right)^{\frac{2}{3}}\right]^{-\frac{3}{4}} = \frac{y}{x}$.

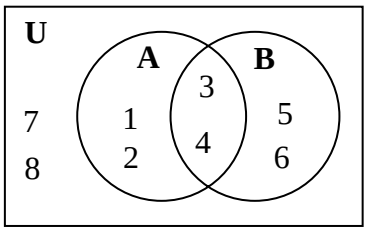
b. Divide $2+3i$ by $1+i$ and separate real and imaginary parts of the answer.

(ATTEMPT EITHER PART a OR PART b OF Q.2.)

Q.2. (Total 4 Marks)

a. If $A=\{1,3,5,7,9\}$ and $B=\{7,9,10,11\}$, then find $(A\cup B)-(A\cap B)$ and $(A\cap B)-(A\cup B)$. (4 Marks)

b. For the given Venn diagram:



i. shade $(A \cup B)'$. (1 Mark)

ii. find $A' \cap B'$. (2 Marks)

iii. compare the results of i and ii and write a relation between $(A \cup B)'$ and $A' \cap B'$. (1 Mark)

Q.3. (Total 3 Marks)

Find the value of x for $\log_4(3x-3) = \frac{1}{2}$.

(ATTEMPT EITHER Q.4. OR Q.5.)

Q.4. (Total 4 Marks)

a. If $a+b+c=16$ and $ab+bc+ca=56$, then find the value of $a^2+b^2+c^2$. (3 Marks)

b. If $x=2$, then find the value of x^3-1. (1 Mark)

(ATTEMPT EITHER Q.4. OR Q.5.)

Q.5. (Total 4 Marks)

Prove that $(a+b)(a-b)\left(a^2+ab+b^2\right)\left(a^2-ab+b^2\right)=a^6-b^6$.

(ATTEMPT EITHER Q.6. OR Q.7.)

Q.6. (Total 5 Marks)

Factorize the following completely:

i. $(a+b)^2 - 49$ (2 Marks)

ii. $20x^2y - 20xy + 5y$ (3 Marks)

(ATTEMPT EITHER Q.6. OR Q.7.)

Q.7. (Total 5 Marks)

Factorize $(x+1)(x+4)(x+2)(x+3) - 24$ completely.

(ATTEMPT EITHER Q.8. OR Q.9.)

Q.8. (Total 4 Marks)

a. If $4 : x :: 2x : 9$, then find the value of x. (2 Marks)

b. The model of an apartment is in the scale 1 : 40. If the apartment is 25 meters high, then find the height of the model apartment. (2 Marks)

(ATTEMPT EITHER Q.8. OR Q.9.)

Q.9. (Total 4 Marks)

Find the value of x for $\frac{\sqrt{x+3}-\sqrt{x-3}}{\sqrt{x+3}+\sqrt{x-3}} = \frac{2}{3}$.

Q.10. (Total 5 Marks)

Prove that the product of matrix $\begin{bmatrix} -3 & 5 \\ -1 & 2 \end{bmatrix}$ and its inverse is equal to unit matrix.

Q.11. (Total 4 Marks)

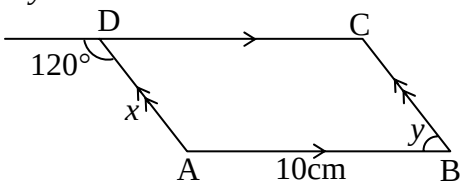
The thickness of 8 randomly selected steel strips is shown below:

Sample	1	2	3	4	5	6	7	8
Thickness (in mm)	17.5	17.6	18.3	18.5	18.5	18.6	19.1	19.3

Find the standard deviation in thickness of the steel strips.

Q.12. (Total 4 Marks)

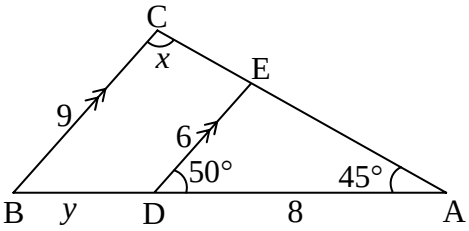
The perimeter of the given parallelogram *ABCD* is 38 cm. Find *x* and *y*.



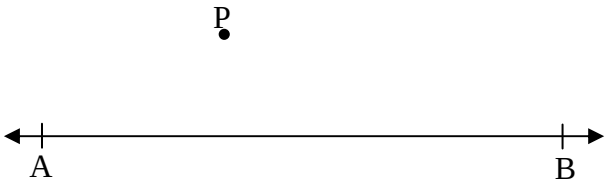
(ATTEMPT EITHER PART a OR PART b OF Q.13.)

Q.13. (Total 4 Marks)

a. In the following diagram *BC* is parallel to *DE*. Find the values of *x* and *y*. (4 Marks)



b. If a point *C* is lying on line *AB*, then construct a line *PC* such that *PC* is the shortest distance between point *P* and line *AB*. Also write measurement of $\overline{PC}$. (4 Marks)



Q.14. (Total 4 Marks)

Construct a triangle *ABC* with $mAB = 6\text{cm}$, $m\angle A = 50°$ and $m\angle B = 60°$. Also draw two medians of the triangle *ABC*.

END OF PAPER

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